

Honeypot Server for Attack Detection

Project	Honeypot Server
Environment	Kali Linux (Oracle VM)
Tools	Cowrie, Fail2Ban, Python, GeoLite2
Objective	Detect and analyze cyber attacks

Introduction

This project implements a low-interaction SSH honeypot using Cowrie to simulate a vulnerable server and capture malicious activities in a controlled environment.

Objectives

- Deploy a honeypot system
- Capture attacker credentials and commands
- Analyze attack patterns
- Block malicious IPs using Fail2Ban
- Visualize attack origins

Implementation

The honeypot was deployed on Kali Linux within a virtual machine. Cowrie was configured on port 2222 to emulate an SSH service. Attack simulations were performed using Hydra to generate brute-force login attempts.

Logging & Monitoring

Cowrie logs all attacker interactions, including IP addresses, login attempts, and executed commands. Logs are stored in both text and JSON formats.

Attack Analysis

Analysis shows that most attacks are automated and target common credentials such as 'root' and 'admin'. Frequently used passwords include '123456' and 'password'.

Intrusion Prevention

Fail2Ban monitors the honeypot logs and bans IP addresses after repeated failed login attempts, improving system defense against brute-force attacks.

Geolocation Visualization

A Python script was used to map attacker IPs to geographic locations using the GeoLite2 database, providing insight into attack origins.

Results

- Successful honeypot deployment
- Real-time attack detection
- Automated IP blocking
- Identification of attack patterns

Conclusion

This project demonstrates the effectiveness of honeypots in cybersecurity research. It provides valuable insights into attacker behavior and enhances defensive strategies.